

CINCINNATI TRANSIT AUTHORITY 2026

FOH CHANNEL PROCESSING — DiGiCo Quantum 225 — NATIVE PROCESSING

Q225 EQ + Dynamics + Dynamic EQ + Mustard Processing + Spice Rack

VENUE: Memorial Hall ('Memo') — Cincinnati, OH — RT60: ~2.2s — Capacity: 556 seats

EQ BANDS: High → Low Order | Thresholds: Calibrated for -25 to -20 dBFS Input Metering

Q225 EQ: 6-band parametric — Bell, Hi/Lo Shelf, HPF, LPF per band. Bands documented High→Low to match console scroll workflow.

Dynamic EQ: Threshold-responsive bands in Q225 — ideal for Memo's 200-400 Hz mud control that responds to playing dynamics. Attack/Release calibrated per instrument.

Mustard Processing — BLUE: Neve-style (warm/musical) | RED: API-style (punchy/gritty) | PURPLE: Optical/LA-2A (smooth/program-dependent) | GREEN: FET/1176 (fast/aggressive).

Spice Rack: Q225 harmonic enhancers and transient shapers — used where noted. Harmonics = odd/even harmonic generation. Transient Shaper = attack/sustain control.

Processing Order: HPF → EQ → Dynamics (Gate then Comp) → Mustard (insert) → Output.

INPUT LIST SUMMARY — Cincinnati Transit Authority 2026

Ch	Instrument	Mic / DI	Mustard	Spice Rack	Dynamic EQ
1	Kick	Earthworks DM6 SeisMic	Red	Yes	2 bands
2	Snare	Lauten Audio Snare Mic (FET Condenser)	Blue	—	2 bands
3	Hi-Hat	AKG C451	—	—	—
4	Rack Tom	Earthworks DM17	Red	Yes	1 band
5	Floor Tom 1	Earthworks DM17	Red	—	1 band
6	Floor Tom 2	Earthworks DM17	Red	—	1 band
7	OH Left	Earthworks SR20	—	—	—
8	OH Right	Earthworks SR20	—	—	—
9	Knee Mic (Shells)	12 Gauge Gold12 (Shotgun)	—	—	—
10	Tracks (Playback)	DI / XLR	—	—	—
11	Bass DI	DI (Active)	Purple	—	2 bands
12	Bass AMP	Earthworks DM6 SeisMic	Red	—	1 band
13	Guitar Left	DI — Pedalboard Direct	Red	Yes	1 band
14	Guitar Right	DI — Pedalboard Direct	Red	Yes	1 band
15	Acoustic Guitar	DI (Active)	Blue	—	1 band
16	Trombone	Shure Beta 98H/C	—	Yes	2 bands
17	Trumpet	Shure Beta 98H/C	—	Yes	2 bands
18	Saxophone	Shure Beta 98H/C	—	Yes	2 bands
19	Flute	DPA 4099 (d:vote CORE)	—	—	1 band
20	Keys 1	DI (Active)	Blue	—	1 band
21	Keys 2	DI (Active)	Blue	—	1 band
22	Keys Vocal	Shure Beta 58	—	—	1 band

23	Bass Vocal	Shure Beta 58	—	—	1 band
24	Drum Vocal	Shure Beta 58	—	—	2 bands
25	Guitar Vocal	Shure Beta 58	—	—	1 band
26	Star Vocal (Lead)	Shure Beta 58	Purple	—	2 bands

Ch 1 — Kick | Earthworks DM6 SeisMic

Q225 EQ — High → Low Band Order

Section	Parameter	Details
EQ — Band 1	LPF	8 kHz, 24dB/oct — DM6 captures full-range; roll off upper content the room smears
EQ — Band 2	Hi Bell	3.5 kHz / +3 dB / Q 2.5 — Beater click and attack; cuts through horn section in the mix
EQ — Band 3	Mid Bell	300 Hz / -5 dB / Q 1.5 — PRIMARY mud cut; Memo RT60 2.2s blooms this range hard on kick
EQ — Band 4	Lo Bell	80 Hz / +3 dB / Q 1.2 — Sub thump and fundamental; Q225 wide bell gives musical body
EQ — Band 5	Lo Shelf	40 Hz / -3 dB — Reduce sub rumble room modes below usable kick freq
EQ — HPF	HPF	60 Hz, 18dB/oct — Hard HPF; Memo's room modes below 60 Hz create sub buildup

Q225 DYNAMICS — Gate + Compressor

Section	Parameter	Details
COMP	Threshold	-22 dBFS — Every hit engages the compressor; calibrated for your -25 to -20 dBFS window
COMP	Ratio	4:1
COMP	Attack	1.5 ms — Fast enough for SSL-style 'smack' on the beater
COMP	Release	50 ms auto — Q225 auto release program-dependent
COMP	Knee	Hard — Kick comp should be decisive
COMP	Makeup	+3 dB
GATE	Threshold	-38 dBFS
GATE	Range	-30 dB — Partial close; preserves natural shell decay
GATE	Attack	0.5 ms — Instant open on beater
GATE	Hold	50 ms
GATE	Release	200 ms

DYNAMIC EQ BANDS — Q225 Nodal / Dynamic EQ

Band A — 300 Hz / Threshold: -28 dBFS / Ratio: 3:1 / Attack: 10 ms / Release: 80 ms / Cut: -4 dB max — Dynamic mud control; engages on every hard kick hit and pulls the 300 Hz bloom before it excites the room's RT60

Band B — 80 Hz / Threshold: -25 dBFS / Ratio: 2:1 / Attack: 5 ms / Release: 60 ms / Boost: +2 dB max — Subtle dynamic sub enhancement on harder hits only

MUSTARD — RED / Vintage (API-style)

RED / API-style — Insert on kick channel. Drive: +4 dB. Adds punchy low-mid harmonic saturation that gives the DM6's clean condenser sound the weight of a dynamic mic. API-style transformer coloration fattens the kick bottom without muddiness. Engage on output stage only.

SPICE RACK — Q225 Transient Shaper

Attack: -2 dB (reduce — DM6 is already transient-rich) | Sustain: +3 dB (enhance — adds tail and room body to the condenser's fast decay) | Use sparingly; Memo's RT60 provides natural sustain.

Engineer Notes: The DM6 SeisMic is a full-range condenser on kick — it captures everything from 20 Hz up, which is a gift and a challenge in a 2.2s reverberant room. The Q225's Dynamic EQ at 300 Hz is the most critical element: a static -5 dB cut cleans the mud, but the Dynamic EQ responds to each hit's intensity and pulls harder when the kick drives harder into the room. Mustard Red gives the API-style transformer saturation that makes the clean DM6 signal feel like a classic kick sound. Gate range of -30 dB (not full close) preserves natural ring.

Ch 2 — Snare | Lauten Audio Snare Mic (FET Condenser)

Q225 EQ — High → Low Band Order

Section	Parameter	Details
EQ — Band 1	LPF	12 kHz, 12dB/oct — Roll off hi-hat shimmer bleed
EQ — Band 2	Hi Bell	10 kHz / +1.5 dB / Q 2.0 — Air and crack; Lauten has a natural dip near 8 kHz
EQ — Band 3	Hi-Mid Bell	5 kHz / +3 dB / Q 2.5 — Stick impact; key presence freq for snare in a dense mix
EQ — Band 4	Lo-Mid Bell	250 Hz / -5 dB / Q 1.5 — Snare ring mud; 200-300 Hz builds in reverberant room on every backbeat
EQ — Band 5	Lo Bell	100 Hz / +1.5 dB / Q 1.0 — Snare body weight
EQ — HPF	HPF	100 Hz, 18dB/oct — Remove kick bleed below snare fundamental

Q225 DYNAMICS — Gate + Compressor

Section	Parameter	Details
COMP	Threshold	-20 dBFS
COMP	Ratio	3:1
COMP	Attack	5 ms — Slow enough to let crack through
COMP	Release	80 ms auto
COMP	Knee	Soft — More musical on snare
COMP	Makeup	+2 dB
GATE	Threshold	-32 dBFS
GATE	Range	-20 dB — Partial; preserves ring between hits
GATE	Attack	0.5 ms
GATE	Hold	30 ms
GATE	Release	150 ms

DYNAMIC EQ BANDS — Q225 Nodal / Dynamic EQ

Band A — 250 Hz / Threshold: -26 dBFS / Ratio: 3:1 / Attack: 8 ms / Release: 60 ms / Cut: -4 dB max — This is the PRIMARY Memo tool on snare; snare ring at 250 Hz blooms in the 2.2s RT60 and clutters the horn section's low fundamentals. DEQ engages only when the snare is struck, pulling the mud reactively.

Band B — 5 kHz / Threshold: -22 dBFS / Ratio: 2:1 / Attack: 3 ms / Release: 40 ms / Boost: +2 dB max — Dynamic presence enhancement on harder hits; crack cuts through more on loud passages automatically.

MUSTARD — BLUE / Classic (Neve-style)

BLUE / Neve-style — Insert on snare. Drive: +2 dB. Neve-style transformer saturation adds warmth and low-mid density to the Lauten FET condenser, giving it more body than a clean preamp chain. Neve harmonic character complements the snare's natural crack beautifully. Don't over-drive; +2 dB is sufficient.

Engineer Notes: The Lauten Snare Mic's FET condenser character differs from a standard SM57 — it has extended response and a natural dip around 3-4 kHz that the 5 kHz boost compensates for. Mustard Blue's Neve-style saturation gives the Lauten the body and warmth that makes it feel authoritative on the backbeat. The Dynamic EQ at 250 Hz is the most important processing element: snare ring in Memo's 2.2s RT60 is the primary source of mix mud in the 200-300 Hz range, and a Dynamic EQ that pulls -4 dB only when the snare fires keeps the mix clean without removing that frequency statically.

Ch 3 — Hi-Hat | AKG C451

Q225 EQ — High → Low Band Order

Section	Parameter	Details
EQ — Band 1	Hi Shelf	10 kHz / -2 dB — C451 famous presence peak; trim to prevent room shimmer spray
EQ — Band 2	Hi Bell	5 kHz / -2 dB / Q 2.0 — 451 harshness zone; metallic edge becomes ugly in reverberant rooms
EQ — Band 3	FLAT	—
EQ — Band 4	FLAT	—
EQ — Band 5	FLAT	—
EQ — HPF	HPF	250 Hz, 24dB/oct — Aggressive; 451 on hat in live setting picks up enormous kick/snare low-end. Cut everything below 250 Hz.

Q225 DYNAMICS — Gate + Compressor

Section	Parameter	Details
COMP	Threshold	-18 dBFS — Light peak control only
COMP	Ratio	2:1
COMP	Attack	10 ms — Don't compress hat attack transient
COMP	Release	60 ms
COMP	Knee	Soft
COMP	Makeup	+1 dB
GATE	Threshold	-28 dBFS
GATE	Range	-40 dB — Hard close; hat is punctuation
GATE	Attack	0.5 ms
GATE	Hold	20 ms
GATE	Release	80 ms

Engineer Notes: The AKG C451 on hi-hat is a supporting detail mic — not a featured element. The 250 Hz HPF is the key move; in Memo's room the bleed from kick and snare into the hat mic creates low-end mush that wastes headroom. The 10 kHz shelf and 5 kHz cut tame the C451's signature presence peak — in a 2.2s room, that shimmer builds into a reverberant top-end shower that obscures the mix. Keep this fader conservative and trust the overheads to carry the cymbal sound.

Ch 4 — Rack Tom | Earthworks DM17

Q225 EQ — High → Low Band Order

Section	Parameter	Details
EQ — Band 1	LPF	10 kHz, 12dB/oct — DM17 is extended; remove upper content that smears in room
EQ — Band 2	Hi Bell	4 kHz / +2.5 dB / Q 2.5 — Attack click and stick definition
EQ — Band 3	Mid Bell	300 Hz / -4 dB / Q 1.5 — Mud control; Memo RT60 blooms toms at 200-350 Hz
EQ — Band 4	Lo Bell	120 Hz / +3 dB / Q 1.0 — Tom body and thump
EQ — Band 5	Lo Shelf	50 Hz / -2 dB — Sub rumble reduction
EQ — HPF	HPF	80 Hz, 18dB/oct

Q225 DYNAMICS — Gate + Compressor

Section	Parameter	Details
COMP	Threshold	-22 dBFS
COMP	Ratio	4:1
COMP	Attack	1.5 ms — Fast attack on tom for Q225 'smack'
COMP	Release	100 ms auto
COMP	Knee	Hard
COMP	Makeup	+3 dB
GATE	Threshold	-35 dBFS
GATE	Range	-30 dB — Partial; preserves ring
GATE	Attack	0.5 ms
GATE	Hold	60 ms
GATE	Release	250 ms — Long; allow natural tom ring to decay

DYNAMIC EQ BANDS — Q225 Nodal / Dynamic EQ

Band A — 300 Hz / Threshold: -28 dBFS / Ratio: 3:1 / Attack: 8 ms / Release: 100 ms / Cut: -3 dB max — Reactive mud control; engages on tom hits only, prevents the 300 Hz bloom from exciting Memo's RT60 between drum fills

MUSTARD — RED / Vintage (API-style)

RED / API-style — Insert on rack tom. Drive: +3 dB. API character gives DM17's clean condenser sound punchy harmonic density. Enhances the natural attack of the tom without adding muddiness.

SPICE RACK — Q225 Transient Shaper

Attack: 0 dB (neutral) | Sustain: +2 dB — Slightly enhance tom sustain; in Memo the room already provides decay so be conservative here.

Engineer Notes: The DM17 is purpose-built for toms — fast transient response and extended low-end capture. In Memo, the 300 Hz range is the enemy: it blooms in the room's long decay and turns the tom section into a low-mid wash during drum fills. The Dynamic EQ at 300 Hz is reactive — it only pulls when the tom fires, which means the cut is perfectly timed to the performance dynamics rather than always removing that frequency statically.

Ch 5 — Floor Tom 1 | Earthworks DM17

Q225 EQ — High → Low Band Order

Section	Parameter	Details
EQ — Band 1	LPF	10 kHz, 12dB/oct
EQ — Band 2	Hi Bell	3.5 kHz / +2.5 dB / Q 2.5 — Attack click
EQ — Band 3	Mid Bell	250 Hz / -4 dB / Q 1.5 — Mud; floor tom builds 200-300 Hz in reverberant rooms
EQ — Band 4	Lo Bell	90 Hz / +3 dB / Q 1.0 — Deep body; lower than rack tom
EQ — Band 5	Lo Shelf	40 Hz / -2 dB
EQ — HPF	HPF	60 Hz, 18dB/oct — Floor tom sits lower; keep more sub

Q225 DYNAMICS — Gate + Compressor

Section	Parameter	Details
COMP	Threshold	-22 dBFS
COMP	Ratio	4:1
COMP	Attack	2 ms
COMP	Release	120 ms auto
COMP	Knee	Hard
COMP	Makeup	+3 dB
GATE	Threshold	-33 dBFS
GATE	Range	-28 dB
GATE	Attack	0.5 ms
GATE	Hold	80 ms
GATE	Release	300 ms — Floor tom ring; allow natural decay

DYNAMIC EQ BANDS — Q225 Nodal / Dynamic EQ

Band A — 250 Hz / Threshold: -28 dBFS / Ratio: 3:1 / Attack: 8 ms / Release: 120 ms / Cut: -3 dB max — Same reactive mud control as rack tom; floor tom slightly lower center frequency

MUSTARD — RED / Vintage (API-style)

RED / API-style — Insert on floor tom 1. Drive: +3 dB. Matched to rack tom Mustard processing for consistent drum section character.

Engineer Notes: Floor Tom 1 mirrors the rack tom approach but tuned deeper — HPF at 60 Hz (vs 80 Hz), LF Bell at 90 Hz (vs 120 Hz). The 250 Hz mud cut is slightly lower than the rack tom to match the floor tom's lower fundamental. Gate release of 300 ms allows the characteristic floor tom ring to sustain naturally before the gate closes. Match levels carefully to the rack tom so the section speaks consistently through fills.

Ch 6 — Floor Tom 2 | Earthworks DM17

Q225 EQ — High → Low Band Order

Section	Parameter	Details
EQ — Band 1	LPF	10 kHz, 12dB/oct
EQ — Band 2	Hi Bell	3.5 kHz / +2 dB / Q 2.5
EQ — Band 3	Mid Bell	240 Hz / -4 dB / Q 1.5 — Deepest mud cut of the tom group
EQ — Band 4	Lo Bell	80 Hz / +3 dB / Q 1.0 — Deepest body boost
EQ — Band 5	Lo Shelf	35 Hz / -2 dB
EQ — HPF	HPF	55 Hz, 18dB/oct — Deepest floor tom; preserve low fundamental

Q225 DYNAMICS — Gate + Compressor

Section	Parameter	Details
COMP	Threshold	-22 dBFS
COMP	Ratio	4:1
COMP	Attack	2 ms
COMP	Release	130 ms auto
COMP	Knee	Hard
COMP	Makeup	+3 dB
GATE	Threshold	-33 dBFS
GATE	Range	-28 dB
GATE	Attack	0.5 ms
GATE	Hold	80 ms
GATE	Release	320 ms

DYNAMIC EQ BANDS — Q225 Nodal / Dynamic EQ

Band A — 240 Hz / Threshold: -28 dBFS / Ratio: 3:1 / Attack: 8 ms / Release: 130 ms / Cut: -3 dB max

MUSTARD — RED / Vintage (API-style)

RED / API-style — Insert on floor tom 2. Drive: +3 dB. Consistent API character across all three toms.

Engineer Notes: Floor Tom 2 is the deepest drum on the kit. HPF at 55 Hz, LF Bell at 80 Hz, mud cut at 240 Hz — each slightly lower than Floor Tom 1. All three toms (Rack, Floor 1, Floor 2) should carry the same Mustard Red character so they sound unified as a section through fills. Gate release of 320 ms is the longest on any channel — this tom has the most natural sustain.

Ch 7 — OH Left | Earthworks SR20

Q225 EQ — High → Low Band Order

Section	Parameter	Details
EQ — Band 1	Hi Shelf	12 kHz / -2 dB — SR20 is extended; tame to prevent room reverb from building harsh top-end wash
EQ — Band 2	Hi Bell	3 kHz / -2 dB / Q 2.0 — SR20 slight presence lift; can sound harsh in reverberant room
EQ — Band 3	Mid Bell	200 Hz / -3 dB / Q 1.2 — Low-mid mud from full kit pickup
EQ — Band 4	FLAT	—
EQ — Band 5	FLAT	—
EQ — HPF	HPF	120 Hz, 18dB/oct — Remove low-end kit rumble; close mics handle kick/snare body

Q225 DYNAMICS — Gate + Compressor

Section	Parameter	Details
COMP	Threshold	-16 dBFS — Gentle; overheads should breathe
COMP	Ratio	2:1
COMP	Attack	15 ms — Slow; preserve cymbal transients
COMP	Release	300 ms auto — Long; keep overheads open
COMP	Knee	Soft
COMP	Makeup	+1 dB
GATE	State	OFF — NEVER gate overheads

Engineer Notes: The SR20 overhead pair are ultra-linear small-diaphragm condensers — they'll tell you exactly what Memo sounds like. No Mustard, no Spice Rack — the SR20 pair should be as transparent as possible. The 120 Hz HPF keeps low-end out of the overheads without losing cymbal body. Critical: keep overhead level conservative in Memo. The room adds natural ambience and the overheads will pick up the room's RT60 aggressively. Phase-check with Smaart v9 against kick and snare before the show.

Ch 8 — OH Right | Earthworks SR20

Q225 EQ — High → Low Band Order

Section	Parameter	Details
EQ — Band 1	Hi Shelf	12 kHz / -2 dB
EQ — Band 2	Hi Bell	3 kHz / -2 dB / Q 2.0
EQ — Band 3	Mid Bell	200 Hz / -3 dB / Q 1.2
EQ — Band 4	FLAT	—
EQ — Band 5	FLAT	—
EQ — HPF	HPF	120 Hz, 18dB/oct

Q225 DYNAMICS — Gate + Compressor

Section	Parameter	Details
COMP	Threshold	-16 dBFS
COMP	Ratio	2:1
COMP	Attack	15 ms
COMP	Release	300 ms auto
COMP	Knee	Soft
COMP	Makeup	+1 dB
GATE	State	OFF — NEVER gate overheads
PHASE	Check	Verify L/R phase alignment in Smaart v9 — SR20 pair must be time-aligned for mono compatibility in Memo

Engineer Notes: OH Right matched exactly to OH Left. The additional priority on this channel is phase alignment — verify the SR20 pair against kick and snare in Smaart v9 before soundcheck begins. Phase cancellation between overheads and close mics is amplified by Memo's room decay and will cause low-end smearing that's nearly impossible to fix during the show.

Ch 9 — Knee Mic (Shells) | 12 Gauge Gold12 (Shotgun)

Q225 EQ — High → Low Band Order

Section	Parameter	Details
EQ — Band 1	LPF	6 kHz, 24dB/oct — Heavy roll-off; this mic's job is body, not detail. Prevents adding room shimmer on top of Memo's RT60.
EQ — Band 2	Hi Shelf	5 kHz / -3 dB — Additional high cut beyond LPF; keep this mic entirely in the low/low-mid range
EQ — Band 3	Mid Bell	300 Hz / -3 dB / Q 1.2 — Mud control even on ambient mic
EQ — Band 4	Lo Bell	100 Hz / +2 dB / Q 1.0 — Shell resonance and room body
EQ — Band 5	FLAT	—
EQ — HPF	HPF	120 Hz, 18dB/oct — Remove floor rumble from shotgun pickup

Q225 DYNAMICS — Gate + Compressor

Section	Parameter	Details
COMP	Threshold	-20 dBFS
COMP	Ratio	3:1
COMP	Attack	20 ms — Ambient mic; let the room breathe through it
COMP	Release	500 ms — Long; glue/blend element
COMP	Knee	Soft
COMP	Makeup	+2 dB
GATE	State	OFF — Ambient mic should remain open
LEVEL	Blend	INTENTIONAL USE: -8 to -12 dB below OH level. Low blend for shell body only. First fader to pull if the kit sounds washy.

Engineer Notes: INTENTIONAL LOW-BLEND USE for shell body and kit size. The 12 Gauge Gold12 at knee level captures the resonant body of the full kit — blend it underneath the close mics and overheads at -8 to -12 dB below OH level. Its job is to add thickness and size, not ambience. The heavy high-end cutting (LPF at 6 kHz, HF shelf at 5 kHz) keeps this mic entirely out of the high-frequency range where Memo's RT60 causes smearing. If the kit starts sounding washy during the show, pull this fader first.

Ch 10 — Tracks (Playback) | DI / XLR

Q225 EQ — High → Low Band Order

Section	Parameter	Details
EQ — Band 1	Hi Shelf	12 kHz / +1 dB — Small air boost; helps digital playback breathe in the room
EQ — Band 2	FLAT	—
EQ — Band 3	Mid Bell	200 Hz / -2 dB / Q 1.2 — Small Memo mud cut on playback low-mid
EQ — Band 4	Lo Shelf	60 Hz / -2 dB — Reduce sub that the room will amplify
EQ — Band 5	FLAT	—
EQ — HPF	HPF	40 Hz, 12dB/oct — Remove subsonic content only

Q225 DYNAMICS — Gate + Compressor

Section	Parameter	Details
COMP	Threshold	-12 dBFS — Light limiting; tracks are already mastered
COMP	Ratio	2:1 — Gentle safety limiter
COMP	Attack	20 ms
COMP	Release	400 ms
COMP	Knee	Soft
COMP	Makeup	0 dB — Tracks set own level
GATE	State	OFF — Never gate playback
NOTE	Stereo	Run as stereo linked pair; tracks are stereo content

Engineer Notes: Playback tracks are already mixed and mastered — minimal processing is the goal. The Q225 provides gentle adaptation to Memo's acoustic environment: a small 200 Hz cut prevents playback low-mids from blooming in the room's decay. Keep the compressor at 2:1 as a safety limiter only. The 40 Hz HPF removes subsonic content that would excite Memo's room modes.

Ch 11 — Bass DI | DI (Active)

Q225 EQ — High → Low Band Order

Section	Parameter	Details
EQ — Band 1	LPF	5 kHz, 12dB/oct — DI bass has full spectrum; roll off upper harshness
EQ — Band 2	Hi Bell	800 Hz / -2.5 dB / Q 2.0 — Bass honk zone; removes midrange mud
EQ — Band 3	Hi-Mid Bell	300 Hz / -4 dB / Q 1.5 — PRIMARY Memo mud cut on bass; low-mids bloom aggressively in 2.2s decay
EQ — Band 4	Lo Bell	80 Hz / +3 dB / Q 1.0 — Bass fundamental; wide Q for musical sub, not peaked
EQ — Band 5	Lo Shelf	40 Hz / -2 dB — Sub room mode control
EQ — HPF	HPF	40 Hz, 12dB/oct

Q225 DYNAMICS — Gate + Compressor

Section	Parameter	Details
COMP	Threshold	-22 dBFS
COMP	Ratio	4:1
COMP	Attack	2 ms — Fast; controls pick attack on DI bass
COMP	Release	80 ms auto
COMP	Knee	Hard
COMP	Makeup	+4 dB
GATE	State	OFF — Bass sustains naturally between notes

DYNAMIC EQ BANDS — Q225 Nodal / Dynamic EQ

Band A — 300 Hz / Threshold: -26 dBFS / Ratio: 3:1 / Attack: 10 ms / Release: 100 ms / Cut: -3 dB max — Reactive mud control; bass low-mids compete directly with the horn section's fundamental range. DEQ ensures the cut responds to playing dynamics — heavier playing = more mud pulled.

Band B — 80 Hz / Threshold: -24 dBFS / Ratio: 2:1 / Attack: 5 ms / Release: 60 ms / Boost: +2 dB max — Dynamic sub enhancement on harder playing passages only; adds low-end weight without constant boominess.

MUSTARD — PURPLE / Optical (LA-2A style)

PURPLE / Optical (LA-2A style) — Insert on bass DI. This is a second compressor stage for attitude and vibe, NOT gain control (the Q225 comp handles that). LA-2A style optical compression adds program-dependent smoothing that glues the bass lines without audible pumping. Program-dependent release is ideal for bass DI. Drive: +2 dB. Threshold: set to just barely engaging on the loudest passages.

Engineer Notes: Bass DI through the Q225 with Mustard Purple (optical) is a classic chain — the Q225 compressor handles gain control with fast attack, and the Mustard Purple LA-2A emulation adds the smooth, program-dependent attitude that makes bass feel locked in. The 300 Hz Dynamic EQ is the critical Memo move: this is where the bass and the horn section compete most directly. A reactive cut that responds to playing dynamics keeps the bass present without clouding the horns. Blend DI at 70% with Amp mic (CH 12) at 30%.

Ch 12 — Bass AMP | Earthworks DM6 SeisMic

Q225 EQ — High → Low Band Order

Section	Parameter	Details
EQ — Band 1	LPF	4 kHz, 24dB/oct — Bass amp is about body; hard roll-off above 4k
EQ — Band 2	Hi Bell	1.2 kHz / +2 dB / Q 1.5 — Amp speaker resonance and cabinet character
EQ — Band 3	Mid Bell	350 Hz / -3.5 dB / Q 1.5 — Amp boxy/boomy mid control in reverberant room
EQ — Band 4	Lo Bell	100 Hz / +3 dB / Q 1.0 — Amp body; slightly higher than DI sub
EQ — Band 5	Lo Shelf	50 Hz / -2 dB
EQ — HPF	HPF	50 Hz, 18dB/oct — Remove subsonic from amp

Q225 DYNAMICS — Gate + Compressor

Section	Parameter	Details
COMP	Threshold	-20 dBFS
COMP	Ratio	3:1
COMP	Attack	8 ms — Amp already compresses naturally; don't over-control
COMP	Release	120 ms auto
COMP	Knee	Soft
COMP	Makeup	+2 dB
GATE	State	OFF
BLEND	DI vs Amp	DI 70% / AMP 30% — DI: clarity and punch; AMP: warmth and character

DYNAMIC EQ BANDS — Q225 Nodal / Dynamic EQ

Band A — 350 Hz / Threshold: -24 dBFS / Ratio: 2:1 / Attack: 12 ms / Release: 120 ms / Cut: -2 dB max — Gentle dynamic amp-boxy control; less aggressive than DI since amp cab already provides some natural filtering

MUSTARD — RED / Vintage (API-style)

RED / API-style — Insert on bass amp. Drive: +4 dB (heavier drive than DI). The DM6 on bass cab gets API-style transformer saturation that reinforces the amp's natural harmonic character. Heavy drive gives it grit and push. This is the 'amp' in the blend — it should sound different and complementary to the Purple optical on the DI.

Engineer Notes: The DM6 on bass amp is an excellent combination — Earthworks designed this mic knowing engineers use it on bass cabs. Its condenser capsule captures cabinet character with extended response a dynamic mic can't match. Mustard Red (API) on the amp channel vs Mustard Purple (optical) on the DI creates a distinctly different character for each source — the blend of the two gives the bass section both control and attitude.

Ch 13 — Guitar Left | DI — Pedalboard Direct

Q225 EQ — High → Low Band Order

Section	Parameter	Details
EQ — Band 1	LPF	5 kHz, 18dB/oct — Simulates speaker cabinet natural rolloff; pedalboard DI lacks this filtering
EQ — Band 2	Hi Shelf	4 kHz / -2 dB — Supplemental presence trimming; pedalboard DI carries sharper spike than mic'd cab
EQ — Band 3	Mid Bell	2.5 kHz / +2 dB / Q 2.0 — Chicago rhythm guitar presence; must be heard under the horn section
EQ — Band 4	Lo-Mid Bell	400 Hz / -3.5 dB / Q 1.5 — Mud/boxiness; critical in Memo's reverberant environment even from pedalboard DI
EQ — Band 5	Lo Bell	200 Hz / +2.5 dB / Q 1.0 — CRITICAL: restore low-end warmth missing without a real guitar cab; without this, pedalboard DI sounds thin and telephonic
EQ — HPF	HPF	100 Hz, 18dB/oct

Q225 DYNAMICS — Gate + Compressor

Section	Parameter	Details
COMP	Threshold	-20 dBFS
COMP	Ratio	4:1 — Even dynamics; guitar must sit under the horn section
COMP	Attack	2 ms — Control pedalboard DI pick attack, sharper than mic'd cab
COMP	Release	100 ms auto
COMP	Knee	Hard
COMP	Makeup	+3 dB
GATE	State	OFF — Pedalboard DI noise floor is typically low

DYNAMIC EQ BANDS — Q225 Nodal / Dynamic EQ

Band A — 400 Hz / Threshold: -24 dBFS / Ratio: 2:1 / Attack: 10 ms / Release: 100 ms / Cut: -2 dB max — Reactive guitar mud control in Memo; 400 Hz guitar boxy-ness blooms in the room on heavier strumming passages

MUSTARD — RED / Vintage (API-style)

RED / API-style — Insert on guitar left. Drive: +4 dB. This is the most important processing on the guitar channels: pedalboard DI has no amp character whatsoever. Mustard Red's API-style transformer saturation adds the harmonic density and grit that a real guitar amp would provide naturally. Heavy drive (+4 dB) is appropriate here.

SPICE RACK — Q225 Transient Shaper

Attack: -1 dB (slightly reduce) | Sustain: +2 dB — Pedalboard DI has sharper transients than mic'd cab; slightly soften attack and add sustain to simulate cab behavior.

Engineer Notes: Pedalboard DI for Guitar Left means the Q225 and Mustard Red must compensate for everything a speaker cabinet normally provides: frequency shaping, harmonic saturation, transient softening, and low-end warmth. The LPF at 5 kHz simulates cab rolloff, the 200 Hz LF Bell restores cab warmth (+2.5 dB is significant — it's the most important EQ move on this channel), and Mustard Red's API saturation provides the amp-like harmonic density that makes pedalboard DI sound like a real guitar in a live context. The Spice Rack transient shaper softens the pick attack to further simulate a mic'd cab response.

Ch 14 — Guitar Right | DI — Pedalboard Direct

Q225 EQ — High → Low Band Order

Section	Parameter	Details
EQ — Band 1	LPF	5 kHz, 18dB/oct — Matched cab simulation to Guitar Left
EQ — Band 2	Hi Shelf	4 kHz / -2 dB
EQ — Band 3	Mid Bell	3 kHz / +2 dB / Q 2.0 — Slightly higher than Guitar Left (2.5 kHz) for stereo upper-mid differentiation
EQ — Band 4	Lo-Mid Bell	400 Hz / -3.5 dB / Q 1.5
EQ — Band 5	Lo Bell	200 Hz / +2.5 dB / Q 1.0 — Matched cab warmth restoration
EQ — HPF	HPF	100 Hz, 18dB/oct

Q225 DYNAMICS — Gate + Compressor

Section	Parameter	Details
COMP	Threshold	-20 dBFS
COMP	Ratio	4:1
COMP	Attack	2 ms
COMP	Release	100 ms auto
COMP	Knee	Hard
COMP	Makeup	+3 dB
GATE	State	OFF
PAN	Position	~2 o'clock — Guitar Left at ~10 o'clock; create width without occupying horn section center

DYNAMIC EQ BANDS — Q225 Nodal / Dynamic EQ

Band A — 400 Hz / Threshold: -24 dBFS / Ratio: 2:1 / Attack: 10 ms / Release: 100 ms / Cut: -2 dB max — Matched reactive mud control to Guitar Left

MUSTARD — RED / Vintage (API-style)

RED / API-style — Insert on guitar right. Drive: +4 dB. Matched to Guitar Left for consistent stereo guitar character. If Guitar Right player runs a driven board vs Guitar Left clean, increase drive to +5 dB and reduce the 200 Hz LF Bell to +1.5 dB.

SPICE RACK — Q225 Transient Shaper

Attack: -1 dB | Sustain: +2 dB — Matched to Guitar Left transient shaper settings.

Engineer Notes: Guitar Right mirrors Guitar Left's full pedalboard DI treatment — LPF cab simulation, 200 Hz warmth restoration, Mustard Red API saturation, transient shaping. The only intentional variation is the HMF center at 3 kHz (vs 2.5 kHz) for stereo upper-mid differentiation. Both guitar channels should feel like a cohesive stereo guitar bed under the horn section, not two separate instruments. Pan L ~10 o'clock, R ~2 o'clock, keep both out of the center where the horns live.

Ch 15 — Acoustic Guitar | DI (Active)

Q225 EQ — High → Low Band Order

Section	Parameter	Details
EQ — Band 1	Hi Shelf	10 kHz / +1.5 dB — Air and sparkle; acoustic needs breath above the horn section
EQ — Band 2	Hi Bell	2.5 kHz / +2 dB / Q 2.5 — String definition and presence; helps acoustic cut through
EQ — Band 3	Mid Bell	350 Hz / -4 dB / Q 1.5 — Acoustic DI boxiness; notorious problem frequency for undersaddle pickups in reverberant rooms
EQ — Band 4	Lo Bell	180 Hz / +1.5 dB / Q 1.0 — Warmth without boom
EQ — Band 5	FLAT	—
EQ — HPF	HPF	120 Hz, 18dB/oct — Remove low-end boominess from acoustic pickup

Q225 DYNAMICS — Gate + Compressor

Section	Parameter	Details
COMP	Threshold	-18 dBFS
COMP	Ratio	3:1
COMP	Attack	8 ms — Let strum attack speak; rhythm is in the transient
COMP	Release	150 ms auto
COMP	Knee	Soft
COMP	Makeup	+2 dB
GATE	State	OFF

DYNAMIC EQ BANDS — Q225 Nodal / Dynamic EQ

Band A — 350 Hz / Threshold: -24 dBFS / Ratio: 2:1 / Attack: 10 ms / Release: 100 ms / Cut: -2 dB max — Acoustic DI boxiness is dynamic; it gets worse on harder strumming. DEQ reacts to playing dynamics for natural-sounding box control.

MUSTARD — BLUE / Classic (Neve-style)

BLUE / Neve-style — Insert on acoustic guitar. Drive: +1 dB (very light). Neve-style harmonic saturation adds warmth to the clinical sound of an undersaddle acoustic pickup without coloring the natural acoustic tone. Very light drive; acoustic guitar should not feel processed.

Engineer Notes: Acoustic guitar DI through Q225 with light Mustard Blue gives the pickup signal warmth without compromising natural character. The 350 Hz cut is the most important move — acoustic DI systems are notorious for this frequency range sounding boxy, and Memo amplifies it. The Dynamic EQ at 350 Hz responds to strumming intensity, pulling harder when the player digs in and releasing during delicate fingerpicking passages. Neve-style saturation (very light) gives the acoustic the warmth a real mic would capture.

Ch 16 — Trombone | Shure Beta 98H/C

Q225 EQ — High → Low Band Order

Section	Parameter	Details
EQ — Band 1	Hi Shelf	10 kHz / -1.5 dB — Trombone can be bright and brassy; tame prevents harshness in Memo's reverb field
EQ — Band 2	Hi Bell	2.5 kHz / +2.5 dB / Q 2.5 — Trombone presence and cut-through; this is where the 'Chicago trombone sound' lives
EQ — Band 3	Mid Bell	300 Hz / -3.5 dB / Q 1.5 — Trombone low-mid bloom in reverberant room; primary mud control
EQ — Band 4	Lo Bell	120 Hz / +2 dB / Q 1.0 — Trombone fundamental warmth and weight
EQ — Band 5	FLAT	—
EQ — HPF	HPF	80 Hz, 18dB/oct — Remove sub rumble; trombone fundamental starts ~75-80 Hz for low Bb

Q225 DYNAMICS — Gate + Compressor

Section	Parameter	Details
COMP	Threshold	-20 dBFS — Control trombone dynamics which vary enormously
COMP	Ratio	3:1
COMP	Attack	8 ms — Brass attack is musical; let it through
COMP	Release	120 ms auto
COMP	Knee	Soft — More musical on brass
COMP	Makeup	+2 dB
GATE	State	OFF — Live brass mics; use HPF to manage bleed

DYNAMIC EQ BANDS — Q225 Nodal / Dynamic EQ

Band A — 300 Hz / Threshold: -24 dBFS / Ratio: 3:1 / Attack: 10 ms / Release: 100 ms / Cut: -3 dB max — Trombone low-mid mud in Memo; this Dynamic EQ is the critical difference between a clear horn section and a muddy one. Engages on every note the trombone plays, pulling the bloom that would otherwise excite Memo's 2.2s decay.

Band B — 2.5 kHz / Threshold: -22 dBFS / Ratio: 2:1 / Attack: 5 ms / Release: 60 ms / Boost: +1.5 dB max — Dynamic presence boost on louder passages; trombone cuts through more on forte sections automatically.

SPICE RACK — Q225 Spice Rack — Harmonics

Engage Odd harmonics enhancer at low level (20-25% drive) — adds subtle warmth and harmonic complexity to the Beta 98H/C's clean capture of trombone brass tone. This simulates the natural harmonic richness of a brass instrument in a way the close mic sometimes fails to capture.

Engineer Notes: Trombone provides the low-frequency foundation of the CTA horn section. The Beta 98H/C is purpose-voiced for brass and handles this extremely well. The Dynamic EQ at 300 Hz is the most important processing element — trombone low-mids in Memo's 2.2s RT60 create the primary mud source in the horn section's blend. The Spice Rack harmonics enhancer (subtle) adds the warmth that the close-mic placement can miss. Keep trombone slightly forward in the horn mix — it's the section's foundation.

Ch 17 — Trumpet | Shure Beta 98H/C

Q225 EQ — High → Low Band Order

Section	Parameter	Details
EQ — Band 1	Hi Shelf	10 kHz / -2 dB — Trumpet is extremely bright; significant taming prevents the room reverb from making it piercing in Memo
EQ — Band 2	Hi Bell	3 kHz / +3 dB / Q 2.5 — Trumpet presence and cut-through; the 'Chicago trumpet sound' lives here. Precise Q important.
EQ — Band 3	Mid Bell	350 Hz / -3 dB / Q 1.5 — Low-mid mud; trumpet in a reverberant room picks up body that clouds the high end
EQ — Band 4	FLAT	— Trumpet fundamental is above 150 Hz; HPF handles the rest
EQ — Band 5	FLAT	—
EQ — HPF	HPF	150 Hz, 18dB/oct — Trumpet fundamental starts ~165 Hz (low E)

Q225 DYNAMICS — Gate + Compressor

Section	Parameter	Details
COMP	Threshold	-18 dBFS — Trumpet players have wildest dynamic range in the band
COMP	Ratio	4:1 — More aggressive than trombone; trumpet dynamics are extreme
COMP	Attack	8 ms — Brass attack is musical; let it through
COMP	Release	100 ms auto
COMP	Knee	Soft
COMP	Makeup	+3 dB
GATE	State	OFF

DYNAMIC EQ BANDS — Q225 Nodal / Dynamic EQ

Band A — 350 Hz / Threshold: -24 dBFS / Ratio: 3:1 / Attack: 8 ms / Release: 80 ms / Cut: -2.5 dB max — Reactive low-mid mud control; same Memo principle as trombone. Trumpet low-mids bloom in the room on forte passages.

Band B — 10 kHz / Threshold: -20 dBFS / Ratio: 2:1 / Attack: 3 ms / Release: 50 ms / Cut: -2 dB max — Dynamic brightness control; prevents trumpet highs from becoming piercing in Memo's reverb field on loud passages only.

SPICE RACK — Q225 Spice Rack — Harmonics

Odd harmonics enhancer at 20% drive — same as trombone treatment. Adds subtle warmth to the Beta 98H/C's clean capture of trumpet brass tone.

Engineer Notes: Trumpet is the most dynamically challenging channel in the show. The 4:1 compression ratio is necessary — trumpet players have enormous dynamic range and in a 556-seat reverberant room, an unrestricted trumpet will dominate the entire mix. The Dynamic EQ at 10 kHz is a Memo-specific tool: it only engages on loud passages and prevents the trumpet's high-frequency content from creating a reverberant shower of bright energy in the room's 2.2s decay. The 3 kHz presence boost is where the Chicago trumpet sound lives — keep it precise.

Ch 18 — Saxophone | Shure Beta 98H/C

Q225 EQ — High → Low Band Order

Section	Parameter	Details
EQ — Band 1	Hi Shelf	8 kHz / -1.5 dB — Sax breathiness and reed noise can be harsh in reverberant rooms
EQ — Band 2	Hi Bell	2 kHz / +2.5 dB / Q 2.0 — Sax presence and note definition; cut-through frequency for sax in the horn section
EQ — Band 3	Mid Bell	300 Hz / -3 dB / Q 1.5 — Sax body mud in reverberant room; consistent Memo treatment across all horns
EQ — Band 4	Lo Bell	150 Hz / +2 dB / Q 1.0 — Sax warmth; especially important for tenor sax
EQ — Band 5	FLAT	—
EQ — HPF	HPF	120 Hz, 18dB/oct — Alto: 150 Hz / Tenor: 100 Hz — adjust if player switches

Q225 DYNAMICS — Gate + Compressor

Section	Parameter	Details
COMP	Threshold	-20 dBFS
COMP	Ratio	3:1
COMP	Attack	8 ms — Sax articulation is in the breath attack; let it through
COMP	Release	120 ms auto
COMP	Knee	Soft
COMP	Makeup	+2 dB
GATE	State	OFF

DYNAMIC EQ BANDS — Q225 Nodal / Dynamic EQ

Band A — 300 Hz / Threshold: -24 dBFS / Ratio: 3:1 / Attack: 10 ms / Release: 100 ms / Cut: -2.5 dB max — Reactive sax mud control; consistent with trombone/trumpet DEQ treatment. Engages on every note, prevents low-mid bloom in Memo's decay.

Band B — 2 kHz / Threshold: -22 dBFS / Ratio: 2:1 / Attack: 5 ms / Release: 60 ms / Boost: +1.5 dB max — Dynamic presence; sax cuts through more on louder passages automatically.

SPICE RACK — Q225 Spice Rack — Harmonics

Odd harmonics enhancer at 20% drive — consistent harmonic enhancement across the full horn section (trombone, trumpet, sax all receive same Spice Rack treatment for a unified horn section character).

Engineer Notes: Saxophone sits between trombone and trumpet in the frequency spectrum. The 2 kHz HMF boost is the sax's defining presence frequency — where sax cuts through a full horn section. The 300 Hz Dynamic EQ is the universal Memo mud-control move for all horn channels. Consistent Spice Rack harmonics treatment across trombone, trumpet, and sax creates a unified horn section character. Note: if the performer switches between alto and tenor, adjust HPF (Alto: 150 Hz, Tenor: 100 Hz) and LF Bell accordingly.

Ch 19 — Flute | DPA 4099 (d:vote CORE)

Q225 EQ — High → Low Band Order

Section	Parameter	Details
EQ — Band 1	Hi Shelf	12 kHz / +1.5 dB — Enhance DPA's natural 10-12 kHz soft boost; flute air and shimmer helps it cut through
EQ — Band 2	Hi Bell	4 kHz / +2 dB / Q 2.5 — Flute note articulation and clarity; where flute 'speaks' in a mix
EQ — Band 3	Mid Bell	400 Hz / -3 dB / Q 1.5 — Flute body/hollow tone buildup in reverberant rooms; DPA's close placement accentuates this
EQ — Band 4	FLAT	— HPF at 200 Hz handles the low end
EQ — Band 5	FLAT	—
EQ — HPF	HPF	200 Hz, 24dB/oct — Aggressive; flute fundamental starts at 261 Hz (middle C). Everything below = breath noise, key noise, floor rumble.

Q225 DYNAMICS — Gate + Compressor

Section	Parameter	Details
COMP	Threshold	-18 dBFS
COMP	Ratio	3:1
COMP	Attack	8 ms — Flute articulation is in the transient; let it speak
COMP	Release	100 ms auto
COMP	Knee	Soft
COMP	Makeup	+2 dB
GATE	State	OFF — DPA 4099 supercardioid rejects well; no gate needed

DYNAMIC EQ BANDS — Q225 Nodal / Dynamic EQ

Band A — 400 Hz / Threshold: -24 dBFS / Ratio: 2:1 / Attack: 10 ms / Release: 80 ms / Cut: -2 dB max — Flute body buildup; engaging only when the player plays harder prevents static removal of the 400 Hz range that's needed for warmth on softer passages.

Engineer Notes: The DPA 4099 is reference-quality — its supercardioid pattern and accuracy make it ideal for flute, which is notoriously difficult to amplify. No Mustard, no Spice Rack — the DPA 4099 should run as clean as possible. The 200 Hz HPF is more aggressive than standard because flute fundamentals start at 261 Hz. Critical performance note: ring out the flute channel last during soundcheck. Gain-before-feedback with the DPA is excellent but Memo's RT60 means the channel will still require careful management. If feedback occurs, check 200-400 Hz first, then 4-5 kHz.

Ch 20 — Keys 1 | DI (Active)

Q225 EQ — High → Low Band Order

Section	Parameter	Details
EQ — Band 1	Hi Shelf	10 kHz / +1.5 dB — Keys air and brilliance
EQ — Band 2	Hi Bell	3 kHz / +1.5 dB / Q 2.0 — Keys presence and attack
EQ — Band 3	Mid Bell	250 Hz / -3 dB / Q 1.5 — Memo mud cut; organ-style sounds especially bloom here
EQ — Band 4	Lo Shelf	80 Hz / -2 dB — Reduce keyboard low-end that will bloom in Memo; organ pads can dominate the room
EQ — Band 5	FLAT	—
EQ — HPF	HPF	50 Hz, 12dB/oct — Remove sub from keys that overwhelms the room

Q225 DYNAMICS — Gate + Compressor

Section	Parameter	Details
COMP	Threshold	-18 dBFS
COMP	Ratio	3:1
COMP	Attack	8 ms — Keys articulation in the front of notes
COMP	Release	150 ms auto
COMP	Knee	Soft
COMP	Makeup	+2 dB
GATE	State	OFF

DYNAMIC EQ BANDS — Q225 Nodal / Dynamic EQ

Band A — 250 Hz / Threshold: -22 dBFS / Ratio: 2:1 / Attack: 15 ms / Release: 150 ms / Cut: -2.5 dB max — Organ/pad sounds build at 250 Hz in Memo's decay; DEQ controls it dynamically so acoustic piano passages don't lose their natural body.

MUSTARD — BLUE / Classic (Neve-style)

BLUE / Neve-style — Insert on Keys 1. Drive: +1.5 dB. Neve-style saturation adds warmth and dimension to keys DI, making digital pianos and organs feel analog. Light drive keeps it musical. Particularly effective on organ patches where transformer coloration enhances the tonewheel character.

Engineer Notes: Keys 1 is the primary keyboard channel — piano, organ, or pad sounds providing the harmonic foundation of the CTA tribute. The 250 Hz Dynamic EQ is critical in Memo: organ and pad sounds at this frequency bloom in the 2.2s decay and create a low-mid wash that obscures the horn section. The DEQ responds to playing intensity — backing off when the player plays softly and pulling harder during loud organ passages. Mustard Blue's Neve warmth is ideal for keyboard DI.

Ch 21 — Keys 2 | DI (Active)

Q225 EQ — High → Low Band Order

Section	Parameter	Details
EQ — Band 1	Hi Shelf	10 kHz / +1 dB
EQ — Band 2	Hi Bell	2.5 kHz / +1.5 dB / Q 2.0 — Slightly different voicing from Keys 1 to avoid spectral masking
EQ — Band 3	Mid Bell	300 Hz / -2.5 dB / Q 1.5 — Mud cut; slightly different center from Keys 1
EQ — Band 4	Lo Shelf	100 Hz / -1.5 dB
EQ — Band 5	FLAT	—
EQ — HPF	HPF	80 Hz, 12dB/oct

Q225 DYNAMICS — Gate + Compressor

Section	Parameter	Details
COMP	Threshold	-18 dBFS
COMP	Ratio	3:1
COMP	Attack	8 ms
COMP	Release	150 ms auto
COMP	Knee	Soft
COMP	Makeup	+2 dB
GATE	State	OFF
NOTE	Confirm	Confirm Keys 2 sound assignments with keyboard player before soundcheck; adjust EQ if Keys 2 runs synth leads vs pads

DYNAMIC EQ BANDS — Q225 Nodal / Dynamic EQ

Band A — 300 Hz / Threshold: -22 dBFS / Ratio: 2:1 / Attack: 15 ms / Release: 150 ms / Cut: -2 dB max — Matched DEQ mud control to Keys 1 with slightly different center frequency for spectral differentiation between the two keyboard channels.

MUSTARD — BLUE / Classic (Neve-style)

BLUE / Neve-style — Insert on Keys 2. Drive: +1.5 dB. Matched to Keys 1 Mustard treatment for consistent keyboard section character. If Keys 2 runs lead synth lines, increase drive to +2.5 dB for more harmonic presence.

Engineer Notes: Keys 2 mirrors Keys 1 with slight EQ variations (HMF at 2.5 kHz vs 3 kHz, mud cut at 300 Hz vs 250 Hz) to prevent spectral masking between the two keyboard channels. The most important pre-soundcheck action: confirm sound assignments with the keyboard player. If Keys 2 runs synth lead lines, the HMF boost should be more aggressive and Mustard drive increased. If Keys 2 runs pads/strings, keep it close to the current settings.

Ch 22 — Keys Vocal | Shure Beta 58

Q225 EQ — High → Low Band Order

Section	Parameter	Details
EQ — Band 1	Hi Shelf	10 kHz / -1 dB — Beta 58 forward presence; small trim prevents harshness in Memo's reverb
EQ — Band 2	Hi Bell	3 kHz / +2 dB / Q 2.5 — Vocal presence; Beta 58 needs nudge at 3k for cut-through in dense mix
EQ — Band 3	Mid Bell	250 Hz / -4 dB / Q 1.5 — CRITICAL Memo vocal mud cut; male vocals build heavily at 200-300 Hz in this room
EQ — Band 4	Lo Bell	180 Hz / +1 dB / Q 1.0 — Small warmth restoration after mud cut
EQ — Band 5	FLAT	—
EQ — HPF	HPF	120 Hz, 18dB/oct — Remove proximity effect and stage rumble

Q225 DYNAMICS — Gate + Compressor

Section	Parameter	Details
COMP	Threshold	-20 dBFS
COMP	Ratio	4:1 — Background vocal needs tight dynamic control
COMP	Attack	6 ms — Let consonant attack through for intelligibility
COMP	Release	100 ms auto
COMP	Knee	Soft
COMP	Makeup	+3 dB
GATE	Threshold	-35 dBFS — Gate reduces keyboard bleed when mic not in use
GATE	Range	-40 dB
GATE	Attack	0.5 ms
GATE	Hold	50 ms
GATE	Release	200 ms

DYNAMIC EQ BANDS — Q225 Nodal / Dynamic EQ

Band A — 250 Hz / Threshold: -26 dBFS / Ratio: 3:1 / Attack: 10 ms / Release: 80 ms / Cut: -3 dB max — Dynamic vocal mud control; Memo RT60 at 250 Hz is the primary intelligibility enemy on vocals. DEQ responds to vocal level, pulling harder when the singer is louder and the room excitation is greatest.

Engineer Notes: The Beta 58 is a proven live vocal workhorse. The 250 Hz LMF cut at -4 dB combined with the Dynamic EQ at the same frequency creates a two-stage approach: static removal of the worst of the mud, with a reactive additional cut when the vocal is loudest and Memo's room excitation is greatest. Gate to clean up keyboard bleed between vocal passages. VCA group all background vocals (Ch 22-25) for consistent level management through the show.

Ch 23 — Bass Vocal | Shure Beta 58

Q225 EQ — High → Low Band Order

Section	Parameter	Details
EQ — Band 1	Hi Shelf	10 kHz / -1 dB
EQ — Band 2	Hi Bell	3 kHz / +2.5 dB / Q 2.5 — Stronger presence boost; bass player may be further from FOH sight-line
EQ — Band 3	Mid Bell	250 Hz / -4.5 dB / Q 1.5 — Slightly more aggressive than Keys Vocal; bass instrument proximity adds extra low-mid buildup in vocal mic
EQ — Band 4	Lo Bell	180 Hz / +1 dB / Q 1.0
EQ — Band 5	FLAT	—
EQ — HPF	HPF	120 Hz, 18dB/oct

Q225 DYNAMICS — Gate + Compressor

Section	Parameter	Details
COMP	Threshold	-20 dBFS
COMP	Ratio	4:1
COMP	Attack	6 ms
COMP	Release	100 ms auto
COMP	Knee	Soft
COMP	Makeup	+3 dB
GATE	Threshold	-33 dBFS
GATE	Range	-40 dB
GATE	Attack	0.5 ms
GATE	Hold	50 ms
GATE	Release	200 ms

DYNAMIC EQ BANDS — Q225 Nodal / Dynamic EQ

Band A — 250 Hz / Threshold: -26 dBFS / Ratio: 3:1 / Attack: 10 ms / Release: 80 ms / Cut: -3 dB max — Matched Dynamic EQ mud control to Keys Vocal; additional half-dB of static cut (-4.5 dB vs -4 dB) compensates for bass instrument proximity effect in the vocal mic.

Engineer Notes: Bass Vocal matches the Keys Vocal template with a slightly more aggressive 250 Hz cut (-4.5 dB vs -4 dB) because bass instrument proximity effect adds extra low-mid energy into the vocal mic. The additional half-dB prevents bass instrument energy from creating intelligibility issues on the vocal. VCA group with all background vocals (Ch 22-25).

Ch 24 — Drum Vocal | Shure Beta 58

Q225 EQ — High → Low Band Order

Section	Parameter	Details
EQ — Band 1	Hi Shelf	10 kHz / -1.5 dB — Drummer's delivery can be energetic; extra high-end taming
EQ — Band 2	Hi Bell	3.5 kHz / +3 dB / Q 2.5 — Strong presence; must overcome drum kit noise pickup
EQ — Band 3	Mid Bell	300 Hz / -5 dB / Q 1.5 — MOST aggressive mud cut of all vocals; drum kit proximity creates significant 200-400 Hz buildup in vocal mic
EQ — Band 4	Lo Bell	200 Hz / +1 dB / Q 1.0
EQ — Band 5	FLAT	—
EQ — HPF	HPF	140 Hz, 24dB/oct — More aggressive HPF; drum kit is the loudest source on stage

Q225 DYNAMICS — Gate + Compressor

Section	Parameter	Details
COMP	Threshold	-18 dBFS — Tighter control; drum vocal mic environment is chaotic
COMP	Ratio	6:1 — Most aggressive ratio of all vocals; drum kit dynamics are enormous
COMP	Attack	3 ms — Fast; control peaks from shouted phrases
COMP	Release	80 ms auto
COMP	Knee	Hard — Decisive compression on this channel
COMP	Makeup	+4 dB
GATE	Threshold	-28 dBFS — Must close hard between vocal phrases to reduce drum bleed
GATE	Range	-50 dB — Very hard close; most aggressive gate in the show
GATE	Attack	0.5 ms
GATE	Hold	80 ms
GATE	Release	250 ms — Long enough to avoid chopping vocal tails

DYNAMIC EQ BANDS — Q225 Nodal / Dynamic EQ

Band A — 300 Hz / Threshold: -22 dBFS / Ratio: 4:1 / Attack: 8 ms / Release: 60 ms / Cut: -4 dB max — Most aggressive vocal DEQ in the show; drum kit proximity at 200-400 Hz is constant and dynamic. This DEQ must respond instantly to prevent kick and snare bleed from appearing in the vocal channel.

Band B — 140 Hz / Threshold: -24 dBFS / Ratio: 3:1 / Attack: 5 ms / Release: 50 ms / Cut: -3 dB max — Additional low-end bleed control; kick drum fundamental bleeds into drum vocal at 100-150 Hz even through the HPF.

Engineer Notes: Drum vocal is one of the most challenging channels in any live mix. The 6:1 compressor ratio, -50 dB gate range, and two-band Dynamic EQ reflect the extreme processing required to make a drum kit vocal intelligible. The Dynamic EQ at 300 Hz at 4:1 ratio is the most aggressive vocal DEQ in the show — it must react instantly to kick and snare bleed. Verify gate settings extensively during soundcheck. If the gate is chewing vocal tails, extend hold time by 20 ms increments until it sounds natural.

Ch 25 — Guitar Vocal | Shure Beta 58

Q225 EQ — High → Low Band Order

Section	Parameter	Details
EQ — Band 1	Hi Shelf	10 kHz / -1 dB
EQ — Band 2	Hi Bell	3 kHz / +2.5 dB / Q 2.5 — Guitar player vocal presence
EQ — Band 3	Mid Bell	250 Hz / -4 dB / Q 1.5 — Standard Memo vocal mud cut
EQ — Band 4	Lo Bell	180 Hz / +1 dB / Q 1.0
EQ — Band 5	FLAT	—
EQ — HPF	HPF	120 Hz, 18dB/oct

Q225 DYNAMICS — Gate + Compressor

Section	Parameter	Details
COMP	Threshold	-20 dBFS
COMP	Ratio	4:1
COMP	Attack	6 ms
COMP	Release	100 ms auto
COMP	Knee	Soft
COMP	Makeup	+3 dB
GATE	Threshold	-33 dBFS
GATE	Range	-40 dB
GATE	Attack	0.5 ms
GATE	Hold	50 ms
GATE	Release	200 ms

DYNAMIC EQ BANDS — Q225 Nodal / Dynamic EQ

Band A — 250 Hz / Threshold: -26 dBFS / Ratio: 3:1 / Attack: 10 ms / Release: 80 ms / Cut: -3 dB max — Matched Dynamic EQ to Keys Vocal and Bass Vocal for consistent background vocal processing.

Engineer Notes: *Guitar Vocal is a standard background/supporting vocal using the same Beta 58 template as Keys Vocal and Bass Vocal. Consistent processing across all background vocal channels (Ch 22-25) creates a cohesive sound for group harmony passages — central to Chicago tribute style. All four background vocals should have identical Dynamic EQ settings so they blend seamlessly when singing together. VCA group Ch 22-25.*

Ch 26 — Star Vocal (Lead) | Shure Beta 58

Q225 EQ — High → Low Band Order

Section	Parameter	Details
EQ — Band 1	Hi Shelf	12 kHz / +1 dB — Lead vocal air and openness; singer should sound forward and present
EQ — Band 2	Hi Bell	3.5 kHz / +3 dB / Q 2.0 — Lead presence; needs to cut through full band + horn section in Memo's acoustic environment
EQ — Band 3	Mid Bell	230 Hz / -5 dB / Q 1.5 — MOST CRITICAL CUT IN THE SHOW; lead vocal at 230 Hz in Memo's 2.2s RT60 will completely wash out if not aggressively controlled
EQ — Band 4	Lo Bell	180 Hz / +1.5 dB / Q 1.0 — Vocal body restored after mud cut
EQ — Band 5	FLAT	—
EQ — HPF	HPF	100 Hz, 18dB/oct — Lead vocal gets looser HPF than backgrounds; more body wanted

Q225 DYNAMICS — Gate + Compressor

Section	Parameter	Details
COMP	Threshold	-22 dBFS — Most sensitive threshold; lead vocal is always in the compressor
COMP	Ratio	3:1 — Moderate ratio; lead vocal dynamics are musical and should not be over-squashed
COMP	Attack	6 ms — Preserve vocal consonant attack for intelligibility
COMP	Release	80 ms auto — Faster than backgrounds; lead vocal should feel energetic and dynamic
COMP	Knee	Soft — Musical compression on lead vocal
COMP	Makeup	+3 dB
GATE	Threshold	-38 dBFS — Looser threshold; lead vocal opens easily
GATE	Range	-20 dB — Partial close ONLY; never hard-gate the lead vocal
GATE	Attack	0.5 ms
GATE	Hold	80 ms
GATE	Release	250 ms

DYNAMIC EQ BANDS — Q225 Nodal / Dynamic EQ

Band A — 230 Hz / Threshold: -28 dBFS / Ratio: 4:1 / Attack: 8 ms / Release: 60 ms / Cut: -4 dB max — THE MOST IMPORTANT DYNAMIC EQ IN THE SHOW. Static -5 dB cut at 230 Hz removes the worst of Memo's vocal mud; Dynamic EQ adds an additional -4 dB reactive cut when the vocal is loudest and the room excitation is at its peak. Combined effect: up to -9 dB of 230 Hz reduction at the loudest moments. This is what makes the lead vocal intelligible in Memo.

Band B — 3.5 kHz / Threshold: -24 dBFS / Ratio: 2:1 / Attack: 5 ms / Release: 50 ms / Boost: +2 dB max — Dynamic presence enhancement; lead vocal cuts through the horn section more on louder phrases automatically. Subtle but audibly effective in a dense ensemble mix.

MUSTARD — PURPLE / Optical (LA-2A style)

PURPLE / Optical (LA-2A style) — Insert on lead vocal. This is second-stage compression for attitude and vibe. The Q225 compressor controls gain; the Mustard Purple optical compression adds the program-dependent smooth response that makes a lead vocal feel like it's glued to the track. LA-2A-style optical release characteristic on vocals is a classic combination. Drive: +2 dB. Set threshold to just barely engaging on moderate phrases — heavy engagement on loud passages only.

Engineer Notes: The lead vocal is the most important channel in the show. The two-stage compressor (Q225 fast compressor + Mustard Purple optical LA-2A) gives both control and attitude. The 230 Hz combined static + Dynamic EQ treatment (up to -9 dB total at peak) is the key to lead vocal intelligibility in Memo's 2.2s RT60 — without it, the room's decay will smear the vocal fundamental into an indistinct wash. The 3.5 kHz Dynamic EQ presence boost ensures the vocal automatically cuts through harder during the most intense musical moments. Keep your right hand on this fader throughout the show. This channel is the show.

GLOBAL FX — Seventh Heaven Pro (Bricasti M7) — Memorial Hall

MEMO-SPECIFIC REVERB PHILOSOPHY: Memorial Hall's 2.2s RT60 means the room IS the reverb. Seventh Heaven Pro should be used conservatively — for dimensional placement, not reverb augmentation. Send levels should be 8-12 dB lower than you would use in a dry room. Decay times should be SHORT — the room adds ~1.5s on top of whatever you dial in.

Bus / Instrument	Parameter	Setting / Details
VOCAL VERB (Lead + BG)	Preset	7H Pro — 'Studio B' or 'Control Room C' — small room, NOT a large hall. The room provides the hall.
VOCAL VERB	Pre-Delay	25-30 ms — Creates separation between dry vocal and reverb tail; critical for intelligibility in Memo
VOCAL VERB	Decay Time	0.8 – 1.0 s — Intentionally short; Memo adds ~1.5s naturally on top
VOCAL VERB	HF Damping	2.5 kHz @ -6 dB — Dark, warm tail; bright reverb tails become harsh in Memo's reflective surfaces
VOCAL VERB	Send Level	-18 to -16 dBFS from vocal channels — Very conservative
HORN VERB (All 4 horns)	Preset	7H Pro — 'Studio C' or 'Wood Room' — warm, tight diffusion enhancing brass without wash
HORN VERB	Pre-Delay	15-20 ms
HORN VERB	Decay Time	0.6 – 0.8 s — Very short; horns in Memo's reverb field are already enveloped by the room
HORN VERB	HF Damping	3 kHz @ -4 dB
HORN VERB	Send Level	-20 to -18 dBFS — For cohesion between horn mics, not effect
KEYS/GTR VERB	Preset	7H Pro — 'Small Stage' — similar to vocal verb, slightly larger room character
KEYS/GTR VERB	Pre-Delay	20 ms
KEYS/GTR VERB	Decay Time	1.0 – 1.2 s
KEYS/GTR VERB	Send Level	-22 dBFS — Minimal; keys and guitar in Memo need very little additional reverb
GLOBAL	DRUMS	DO NOT send drum channels to reverb in Memo. The 2.2s RT60 provides all drum ambience needed. Adding reverb will make drums sound like a cavern.
GLOBAL	Smaart Check	Monitor wet/dry balance in mono with Smaart v9 active to verify reverb tails are not building room mode resonances

CRITICAL NOTES — Memorial Hall • Cincinnati Transit Authority 2026

MEMO RT60 — THE FUNDAMENTAL RULE

Memo's 2.2s RT60 drives every decision in this document: every mud cut, every aggressive HPF, every conservative reverb send, every Dynamic EQ threshold. If your instinct says 'add a bit of reverb' or 'boost the low-mids a touch' — resist it. The room is already doing those things. Your job is to be surgical where the room is generous.

DYNAMIC EQ — WHY IT MATTERS IN THIS ROOM

Static EQ cuts the same amount regardless of performance dynamics. In Memo's reverberant environment, the problem is not constant — it's worst when musicians play hardest. Dynamic EQ that responds to playing level means: quiet passages preserve natural tone, loud passages get the aggressive mud cuts they need. Every DEQ threshold in this document is calibrated for the -25 to -20 dBFS input window. If input levels shift, revisit DEQ thresholds first.

MUSTARD FLAVOR SUMMARY

RED (API) on: Kick, Toms, Bass AMP, Guitars — punchy, gritty, transformer saturation for rhythm section. PURPLE (Optical) on: Bass DI, Lead Vocal — smooth program-dependent compression for musical glue. BLUE (Neve) on: Snare, Acoustic Guitar, Keys 1, Keys 2 — warm, musical, wide Q harmonic richness. NO Mustard on: OH L/R, Hat, Flute, Horns, Tracks, Knee Mic — these sources need transparency.

SPICE RACK USAGE

Spice Rack Harmonics Enhancer (odd harmonics) on: Trombone, Trumpet, Saxophone — consistent treatment creates unified horn section character. Level: 20-25% drive only — subtle enhancement, not effect. Spice Rack Transient Shaper on: Kick (sustain +3), Guitar L/R (attack -1, sustain +2). These are the only Spice Rack applications warranted on this show.

VCA / DCA GROUPING

Group 1 DRUMS: Ch 1-9 | Group 2 BASS: Ch 11-12 (linked) | Group 3 GUITARS: Ch 13-15 | Group 4 HORNS: Ch 16-19 | Group 5 KEYS: Ch 20-21 | Group 6 BG VOCALS: Ch 22-25 | Group 7 LEAD VOCAL: Ch 26 (solo fader) | Group 8 TRACKS: Ch 10

SOUNDCHECK ORDER

1) Kick + Snare — establish low-end floor and Dynamic EQ response. 2) OH L/R — check phase in Smaart v9 against kick and snare. 3) Toms + Hat + Knee Mic blend. 4) Bass DI + AMP blend (DI 70% / AMP 30%). 5) Guitars — verify pedalboard DI warmth from Mustard Red + 200 Hz LF Bell. 6) Full horn section together — check 300 Hz DEQ response as a group. 7) Keys in context of full band. 8) Background vocals — verify gate behavior on drum vocal. 9) Lead vocal LAST — always dial in with the full band playing, never in isolation.

SMAART V9 CHECKPOINTS

Pre-show: Pink noise sweep to identify Memo's room modes (expect nodes ~63 Hz, 125 Hz, 250-315 Hz). Drum soundcheck: OH L/R phase alignment against kick and snare — verify mono compatibility. Full band RTA: verify 200-400 Hz mud control is working across the whole mix. Lead vocal intelligibility: STI target >0.60 in Memo. Wet/dry check: confirm reverb tails not reinforcing room modes.

HORN SECTION — UNIFIED PROCESSING

Trombone (Ch16), Trumpet (Ch17), Sax (Ch18) all share: same 300-350 Hz Dynamic EQ approach, same Spice Rack harmonics treatment. This is intentional — a unified processing character makes the horn section blend as one instrument. During soundcheck bring all four horns up together after individual ringtone and check the blend as a section. Flute (Ch19) is treated separately — no Spice Rack, no Mustard, maximum transparency.

CTA 2026 — DiGiCo Q225 Native Processing — Q225 EQ / Dynamic EQ / Mustard / Spice Rack — Memorial Hall (Memo), Cincinnati OH — RT60 ~2.2s
— All thresholds calibrated for -25 to -20 dBFS input metering.